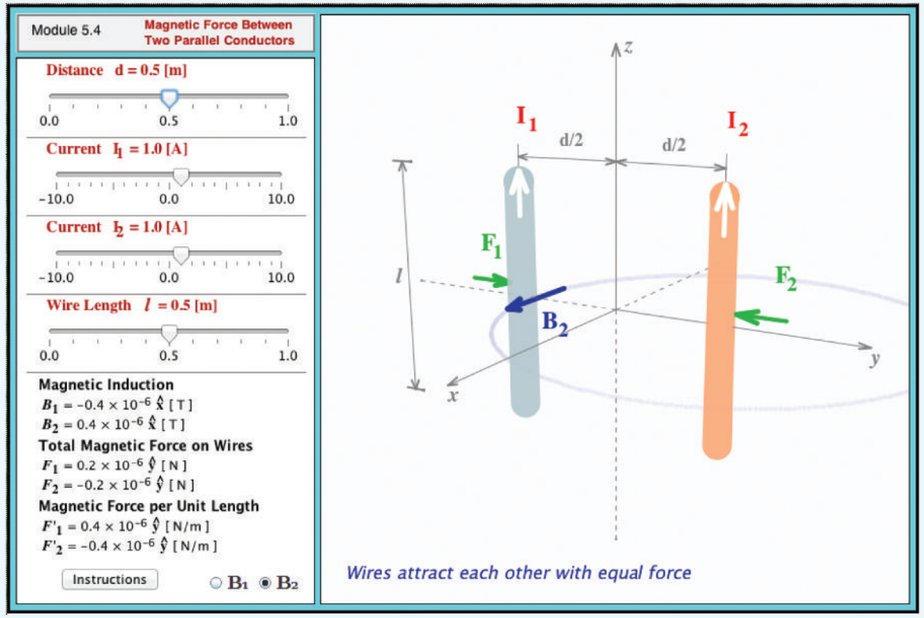
Module 5.4 Magnetic Force between Two Parallel Conductors Observe the direction and magnitude of the force exerted on parallel current-carrying wires.



5-2.3 Magnetic Force Between Two Parallel Conductors

In Section 5-1.1, we examined the magnetic force $\mathbf{F}_m$ that acts on a current-carrying conductor when placed in an external magnetic field. The current in the conductor, however, also generates its own magnetic field. Hence, if two current-carrying conductors are placed in each other's vicinity, each will exert a magnetic force on the other. Let us consider two very long (or effectively infinitely long), straight, free-standing, parallel wires separated by a distance d and carrying currents I_1 and I_2 in the z direction (**Fig. 5-14**) at $y = -d/2$ and $y = d/2$, respectively. We denote by $\mathbf{B}_1$ the magnetic field due to current I_1, defined at the location of the wire carrying current I_2 and, conversely, by $\mathbf{B}_2$ the field due to I_2 at the location of the wire carrying current I_1. From Eq. (5.30), with $I = I_1$, $r = d$, and $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}}$ at the location of I_2, the field $\mathbf{B}_1$ is

$$\mathbf{B}_1 = -\hat{\mathbf{x}}\,\frac{\mu_0 I_1}{2\pi d}. \tag{5.39}$$

The force $\mathbf{F}_2$ exerted on a length l of wire I_2 due to its presence in field $\mathbf{B}_1$ may be obtained by applying Eq. (5.12):

$$\mathbf{F}_2 = I_2 l\hat{\mathbf{z}} \times \mathbf{B}_1 = I_2 l\hat{\mathbf{z}} \times (-\hat{\mathbf{x}})\,\frac{\mu_0 I_1}{2\pi d} = -\hat{\mathbf{y}}\,\frac{\mu_0 I_1 I_2 l}{2\pi d}, \tag{5.40}$$

and the corresponding force per unit length is

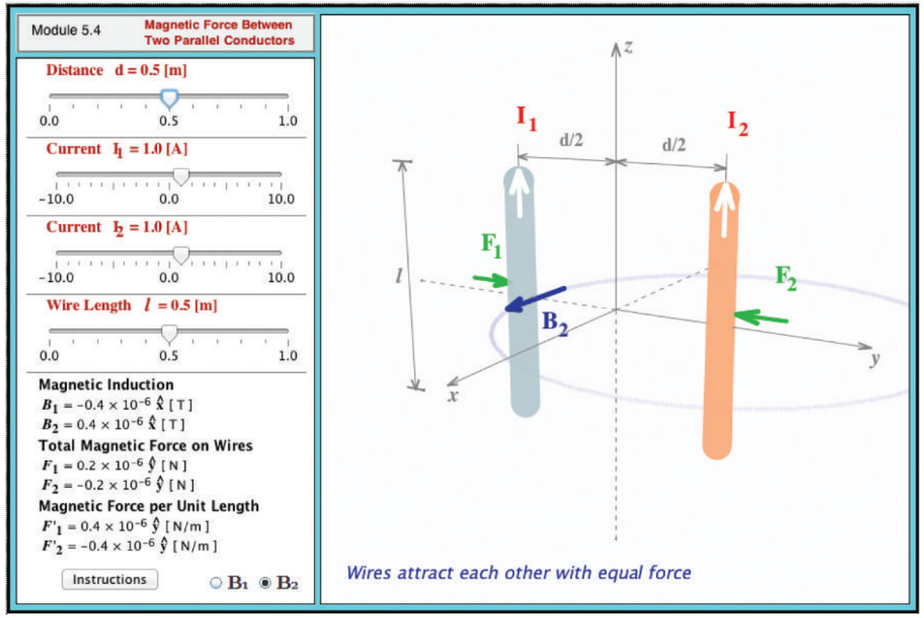
$$\mathbf{F}_2' = \frac{\mathbf{F}_2}{l} = -\hat{\mathbf{y}}\,\frac{\mu_0 I_1 I_2}{2\pi d}. \tag{5.41}$$

A similar analysis performed for the force per unit length exerted on the wire carrying I_1 leads to

$$\mathbf{F}_1' = \hat{\mathbf{y}}\,\frac{\mu_0 I_1 I_2}{2\pi d}. \tag{5.42}$$

► Thus, two parallel wires carrying currents in the same direction attract each other with equal force. If the currents are in opposite directions, the wires would repel one another with equal force. ◄

模块 5.4 两个平行导体之间的磁力 观察施加在平行载流线上的力的方向和大小。



5-2.3 两平行导体之间的磁力

在第 5-1.1 节中，我们研究了当放置在外部磁场中时作用在载流导体上的磁力 F_m。然而，导体中的电流也会产生自己的磁场。因此，如果两个载流导体放置在彼此附近，每个导体都会对另一个导体施加磁力。让我们考虑两条非常长（或实际上无限长）、直的、独立的、平行的导线，其间隔距离 d，并在 z 方向（图 5-14）分别在 y = −d/2 和 y = d/2 处承载电流 I_1 和 I_2。我们用 B_1 表示电流 I_1 产生的磁场，定义在承载电流 I_2 的导线位置，相反，用 B_2 表示在承载电流 I_1 的导线位置处由 I_2 产生的磁场。从方程。 (5.30)，当 $I = I_1$、$r = d$ 和 $\hat{\boldsymbol{\phi}} = -\hat{x}$ 在 I_2 位置时，域 B_1 为

$$\mathbf{B}_1 = -\hat{\mathbf{x}}\,\frac{\mu_0 I_1}{2\pi d}. \tag{5.39}$$

由于存在于磁场 B_1 中，施加在长度为 l 的导线 I_2 上的力 F_2 可以通过应用等式获得： (5.12) ：

$$\mathbf{F}_2 = I_2 l\hat{\mathbf{z}} \times \mathbf{B}_1 = I_2 l\hat{\mathbf{z}} \times (-\hat{\mathbf{x}})\,\frac{\mu_0 I_1}{2\pi d} = -\hat{\mathbf{y}}\,\frac{\mu_0 I_1 I_2 l}{2\pi d}, \tag{5.40}$$

单位长度对应的力为

$$\mathbf{F}_2' = \frac{\mathbf{F}_2}{l} = -\hat{\mathbf{y}}\,\frac{\mu_0 I_1 I_2}{2\pi d}. \tag{5.41}$$

对施加在承载 I_1 的导线上的每单位长度的力进行类似的分析，得出

$$\mathbf{F}_1' = \hat{\mathbf{y}}\,\frac{\mu_0 I_1 I_2}{2\pi d}. \tag{5.42}$$

► 因此，沿相同方向承载电流的两条平行导线以相等的力相互吸引。如果电流方向相反，电线将以相等的力相互排斥。 ◄

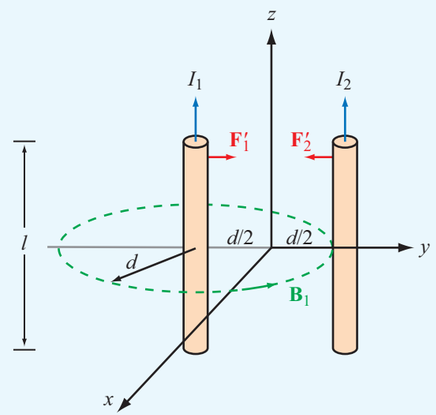


Figure 5-14 Magnetic forces on parallel current-carrying conductors.

that is enclosed by a surface S and contains a total charge $Q = \int_v \rho_v \, dv$ (Section 4-4).

The magnetostatic counterpart of Eq. (5.43), often called ***Gauss's law for magnetism***, is

$$\nabla \cdot \mathbf{B} = 0 \quad \leftrightarrow \quad \oint_S \mathbf{B} \cdot d\mathbf{s} = 0. \tag{5.44}$$

The differential form is one of Maxwell's four equations, and the integral form is obtained with the help of the divergence theorem. Note that the right-hand side of Gauss's law for magnetism is zero, reflecting the fact that the magnetic equivalence of an electric point charge does not exist in nature.

► The hypothetical magnetic analogue to an electric point charge is called a ***magnetic monopole***. Magnetic monopoles, however, always occur in pairs (that is, as dipoles). ◄

No matter how many times a permanent magnet is subdivided, each new piece will always have a north and a south pole, even if the process were to be continued down to the atomic level. Consequently, there is no magnetic equivalence to an electric charge q or charge density ρ_v.

Formally, the name "Gauss's law" refers to the electric case, even when no specific reference to electricity is indicated. *The property described by Eq. (5.44) has been called "the law of nonexistence of isolated monopoles," "the law of conservation of magnetic flux," and "Gauss's law for magnetism," among others.* We prefer the last of the three cited names because it reminds us of the parallelism, as well as the differences, between the electric and magnetic laws of nature.

The difference between Gauss's law for electricity and its magnetic counterpart can be elucidated in terms of field lines. Electric field lines originate from positive electric charges and terminate on negative ones. Hence, for the electric field lines of the electric dipole shown in **Fig. 5-15(a)**, the electric flux through a closed surface surrounding one of the charges is nonzero. In contrast, *magnetic field lines always form continuous closed loops.* As we saw in Section 5-2, the magnetic field lines due to currents do not begin or end at any point; this is true for the linear conductor of **Fig. 5-11** and the circular loop of **Fig. 5-12**, as well as for any current distribution. It is also true for a bar magnet (**Fig. 5-15(b)**). Because the magnetic field lines form closed loops, the net magnetic flux through any closed surface surrounding the south pole of the magnet (or through any other closed surface) is always zero, regardless of its shape.

Exercise 5-9: Suppose that the conductor carrying I_2 in **Fig. 5-14** is rotated so that it is parallel to the x axis. What would $\mathbf{F}_2$ be in that case?

Answer: $\mathbf{F}_2 = 0$. (See EM.)

5-3 Maxwell's Magnetostatic Equations

Thus far, we have introduced the Biot–Savart law for finding the magnetic flux density **B** and field **H** due to any distribution of electric currents in free space, and we examined how magnetic fields can exert magnetic forces on moving charged particles and current-carrying conductors. We now examine two additional important properties of magnetostatic fields.

5-3.1 Gauss's Law for Magnetism

In Chapter 4, we learned that the net outward flux of the electric flux density **D** through a closed surface equals the enclosed net charge Q. We referred to this property as Gauss's law (for electricity) and expressed it mathematically in differential and integral forms as

$$\nabla \cdot \mathbf{D} = \rho_v \quad \leftrightarrow \quad \oint_S \mathbf{D} \cdot d\mathbf{s} = Q. \tag{5.43}$$

Conversion from differential to integral form was accomplished by applying the divergence theorem to a volume v

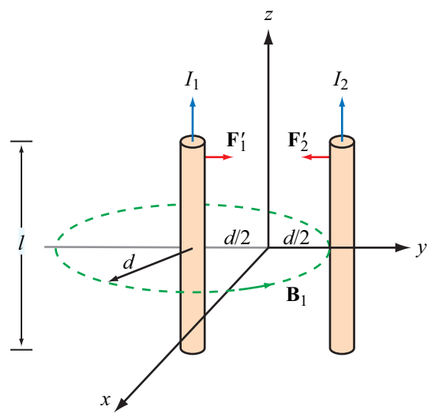


图 5-14 平行载流导体上的磁力。

被表面 S 包围并包含总电荷 $Q = \int_v \rho_v \, dv$ (第 4-4 节)。

等式的静磁对应物。(5.43)，通常称为高斯磁定律，是

$$\nabla \cdot \mathbf{B} = 0 \quad \leftrightarrow \quad \oint_S \mathbf{B} \cdot d\mathbf{s} = 0. \tag{5.44}$$

微分形式是麦克斯韦四个方程之一，积分形式是借助散度定理得到的。请注意，高斯磁性定律的右侧为零，反映了自然界中不存在点电荷的磁性等价这一事实。

► 点电荷的假设磁模拟称为磁单极子。然而，磁单极子总是成对出现（即偶极子）。 ◄

练习5-9：假设图5-14中带有I_2的导体被旋转，使其平行于x轴。在这种情况下 F_2 会是什么？

答案：$F_2 = 0$。（参见EM。）

5-3 麦克斯韦静磁方程

到目前为止，我们已经介绍了毕奥-萨伐尔定律，用于计算自由空间中任何电流分布引起的磁通密度 B 和磁场 H，并且我们研究了磁场如何对移动的带电粒子和载流导体施加磁力。我们现在研究静磁场的另外两个重要特性。

5-3.1 高斯磁定律

在第 4 章中，我们了解到通过封闭表面的电通量密度 D 的净向外通量等于封闭的净电荷 Q。我们将这一性质称为高斯定律（对于电），并以微分和积分形式在数学上将其表示为

$$\nabla \cdot \mathbf{D} = \rho_v \quad \leftrightarrow \quad \oint_S \mathbf{D} \cdot d\mathbf{s} = Q. \tag{5.43}$$

从微分形式到积分形式的转换是通过将散度定理应用于体积 υ 来完成的

无论永磁体被细分多少次，每个新块都将始终有一个北极和一个南极，即使这个过程要继续到原子水平。因此，不存在与电荷 q 或电荷密度 ρ_v 的磁等价。

从形式上讲，“高斯定律”这个名称指的是电力情况，即使没有具体指出电力。方程描述的属性。 (5.44) 被称为“孤立单极子不存在定律”、“磁通量守恒定律”和“高斯磁定律”等。我们更喜欢引用的三个名称中的最后一个，因为它提醒我们自然电定律和磁定律之间的相似性和差异。

高斯电定律和磁定律之间的差异可以通过场线来阐明。电场线源自正电荷并终止于负电荷。因此，对于图 5-15(a) 所示的电偶极子的电场线，通过围绕一个电荷的闭合表面的电通量不为零。相反，磁场线总是形成连续的闭环。正如我们在第 5-2 节中看到的，电流产生的磁力线不会在任何点开始或结束；对于图 5-11 的线性导体和图 5-12 的圆形回路以及任何电流分布都是如此。对于条形磁铁也是如此（图 5-15(b)）。由于磁力线形成闭环，因此通过围绕磁体南极的任何闭合表面（或通过任何其他闭合表面）的净磁通量始终为零，无论其形状如何。

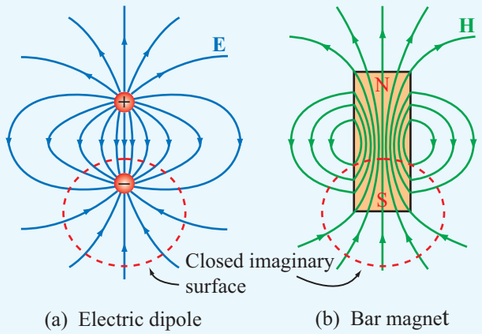


Figure 5-15 Whereas (a) the net electric flux through a closed surface surrounding a charge is not zero, and (b) the net magnetic flux through a closed surface surrounding one of the poles of a magnet is zero.

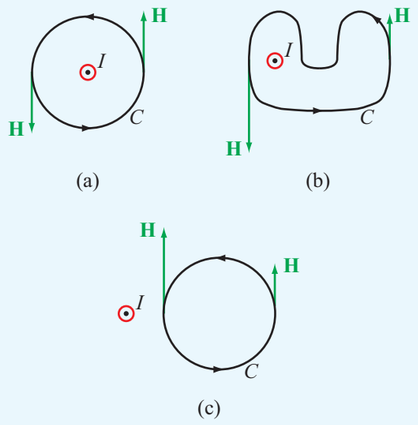


Figure 5-16 Ampère's law states that the line integral of **H** around a closed contour *C* is equal to the current traversing the surface bounded by the contour. This is true for contours (a) and (b), but the line integral of **H** is zero for the contour in (c) because the current *I* (denoted by the symbol ⊙) is not enclosed by the contour *C*.

5-3.2 Ampère's Law

In Chapter 4 we learned that the electrostatic field is conservative, meaning that its line integral along a closed contour always vanishes. This property of the electrostatic field was expressed in differential and integral forms as

$$\nabla \times \mathbf{E} = 0 \quad \Longleftrightarrow \quad \oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = 0. \tag{5.45}$$

Conversion of the differential to integral form was accomplished by applying Stokes's theorem to a surface *S* with contour *C*.

The magnetostatic counterpart of Eq. (5.45), known as ***Ampère's law***, is

$$\nabla \times \mathbf{H} = \mathbf{J} \quad \Longleftrightarrow \quad \oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I, \tag{5.46}$$

where *I* is the total current passing through *S*. The differential form again is one of Maxwell's equations, and the integral form is obtained by integrating both sides of Eq. (5.46) over an open surface *S*,

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{s} = \int_S \mathbf{J} \cdot d\mathbf{s}, \tag{5.47}$$

and then invoking Stokes's theorem with $I = \int \mathbf{J} \cdot d\mathbf{s}$.

► The sign convention for the direction of the contour path *C* in Ampère's law is taken so that *I* and **H** satisfy the right-hand rule defined earlier in connection with the Biot–Savart law. That is, if the direction of *I* is aligned with the direction of the thumb of the right hand, then the direction of the contour *C* should be chosen along that of the other four fingers. ◄

In words, *Ampère's circuital law states that the line integral of* **H** *around a closed path is equal to the current traversing the surface bounded by that path.* To apply Ampère's law, the current must flow through a closed path. By way of illustration, for both configurations shown in **Figs. 5-16(a)** and **(b)**, the line integral of **H** is equal to the current *I*, even though the paths have very different shapes and the magnitude of **H** is not uniform along the path of the configuration in part (b). By the same token, because path (c) in **Fig. 5-16** does not enclose the current *I*, the line integral of **H** along it vanishes, even though **H** is not zero along the path.

When we examined Gauss's law in Section 4-4, we discovered that in practice its usefulness for calculating the electric flux density **D** is limited to charge distributions that possess a certain degree of symmetry and that the calculation procedure

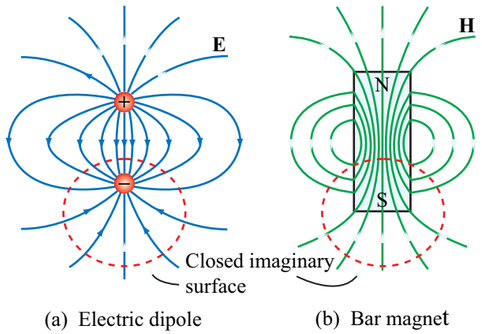


图 5-15 而 (a) 通过围绕电荷的封闭表面的净电通量不为零，并且 (b) 通过围绕磁体之一极的封闭表面的净磁通量为零。

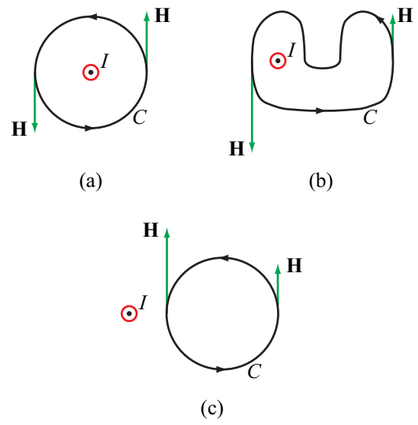


图 5-16 安培环路定律 指出，H 围绕闭合轮廓 C 的线积分等于穿过由轮廓界定的表面的电流。对于轮廓 (a) 和 (b) 来说是这样，但对于 (c) 中的轮廓，H 的线积分为零，因为电流 I（用符号 ⊙ 表示）没有被轮廓 C 包围。

5-3.2 安培环路定律

在第四章中，我们了解到静电场是保守的，这意味着它沿着闭合轮廓的线积分总是消失。静电场的这种性质可以用微分和积分形式表示为

$$\nabla \times \mathbf{E} = 0 \quad \Longleftrightarrow \quad \oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = 0. \tag{5.45}$$

通过将斯托克斯定理应用于具有轮廓 C 的曲面 S 来完成微分形式到积分形式的转换。

等式的静磁对应物。 (5.45)，称为安培环路定律，是

$$\nabla \times \mathbf{H} = \mathbf{J} \quad \Longleftrightarrow \quad \oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = I, \tag{5.46}$$

其中I是通过S的总电流。微分形式又是麦克斯韦方程组之一，积分形式是通过对方程两边积分得到。 (5.46) 在开放表面 S 上，

$$\int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{s} = \int_S \mathbf{J} \cdot d\mathbf{s}, \tag{5.47}$$

然后用 $I = \int J \cdot ds$ 调用斯托克斯定理。

► 轮廓路径 C 方向的符号约定（以 Amp 为单位）采用埃尔定律，使得 I 和 H 满足先前定义的与毕奥-萨伐尔定律相关的右手定则。也就是说，如果I的方向与右手大拇指的方向一致，则轮廓C的方向应沿着其他四个手指的方向选择。◄

用言语来说，放大器埃雷环路定律指出，H 围绕闭合路径的线积分等于流经该路径所界定的表面的电流。应用放大器根据定律，电流必须流经闭合路径。作为说明，对于图 1 和 2 中所示的两种配置，如图 5-16(a) 和 (b) 所示，H 的线积分等于电流 I，尽管路径的形状非常不同，并且 H 的大小沿着 (b) 部分的配置路径并不均匀。出于同样的原因，由于图 5-16 中的路径 (c) 不包含电流 I，因此即使 H 沿路径不为零，H 沿其的线积分也会消失。

当我们研究第 4-4 节中的高斯定律时，我们发现实际上它计算电通量密度 D 的有用性仅限于具有一定程度对称性的电荷分布，并且计算过程

is subject to the proper choice of a Gaussian surface enclosing the charges. A similar restriction applies to Ampère's law: its usefulness is limited to symmetric current distributions that allow the choice of convenient ***Ampèrian contours*** around them, as illustrated by Examples 5-4 to 5-6.

Example 5-4: Magnetic Field of a Long Wire

A long (practically infinite) straight wire of radius a carries a steady current I that is uniformly distributed over its cross section. Determine the magnetic field $\mathbf{H}$ a distance r from the wire axis for (a) $r \le a$ (inside the wire) and (b) $r \ge a$ (outside the wire).

Solution: **(a)** We choose I to be along the $+z$ direction (**Fig. 5-17(a)**). To determine $\mathbf{H}_1 = \mathbf{H}$ at a distance $r = r_1 \le a$, we choose the Ampèrian contour C_1 to be a circular path of radius $r = r_1$ (**Fig. 5-17(b)**). In this case, Ampère's law takes the form

$$\oint_{C_1} \mathbf{H}_1 \cdot d\mathbf{l}_1 = I_1, \tag{5.48}$$

where I_1 is the fraction of the total current I flowing through C_1. From symmetry, $\mathbf{H}_1$ must be constant in magnitude and parallel to the contour at any point along the path. Furthermore, to satisfy the right-hand rule and given that I is along the z direction, $\mathbf{H}_1$ must be in the $+\phi$ direction. Hence, $\mathbf{H}_1 = \hat{\boldsymbol{\phi}} H_1$, $d\mathbf{l}_1 = \hat{\boldsymbol{\phi}} r_1\, d\phi$, and the left-hand side of Eq. (5.48) becomes

$$\oint_{C_1} \mathbf{H}_1 \cdot d\mathbf{l}_1 = \int_0^{2\pi} H_1 (\hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}}) r_1\, d\phi = 2\pi r_1 H_1.$$

The current I_1 flowing through the area enclosed by C_1 is equal to the total current I multiplied by the ratio of the area enclosed by C_1 to the total cross-sectional area of the wire:

$$I_1 = \left(\frac{\pi r_1^2}{\pi a^2}\right) I = \left(\frac{r_1}{a}\right)^2 I.$$

Equating both sides of Eq. (5.48) and then solving for $\mathbf{H}_1$ yields

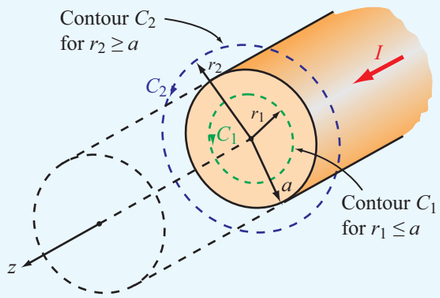
$$\mathbf{H}_1 = \hat{\boldsymbol{\phi}} H_1 = \hat{\boldsymbol{\phi}} \frac{r_1}{2\pi a^2} I \qquad (\text{for } r_1 \le a). \tag{5.49a}$$

(b) For $r = r_2 \ge a$, we choose path C_2, which encloses all the current I. Hence, $\mathbf{H}_2 = \hat{\boldsymbol{\phi}} H_2$, $d\ell_2 = \hat{\boldsymbol{\phi}} r_2\, d\phi$, and

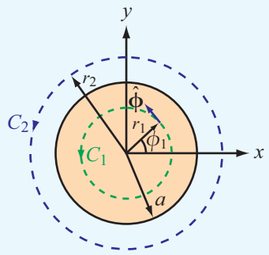
$$\oint_{C_2} \mathbf{H}_2 \cdot d\mathbf{l}_2 = 2\pi r_2 H_2 = I,$$

which yields

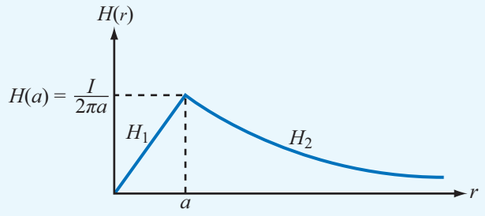
$$\mathbf{H}_2 = \hat{\boldsymbol{\phi}} H_2 = \hat{\boldsymbol{\phi}} \frac{I}{2\pi r_2} \qquad (\text{for } r_2 \ge a). \tag{5.49b}$$



(a) Cylindrical wire



(b) Wire cross section



(c)

Figure 5-17 Infinitely long wire of radius a carrying a uniform current I along the $+z$ direction: (a) general configuration showing contours C_1 and C_2; (b) cross-sectional view; and (c) a plot of H versus r (Example 5-4).

Ignoring the subscript 2, we observe that Eq. (5.49b) provides the same expression for $\mathbf{B} = \mu_0 \mathbf{H}$ as Eq. (5.30), which was derived on the basis of the Biot–Savart law.

The variation of the magnitude of H as a function of r is

（续上页）这取决于能否选取适当的、包围电荷的高斯面。安培环路定律也有类似限制：只有当电流分布具有足够对称性、能够选取方便的安培环路时，它才便于直接求场，如例5-4至5-6所示。

例 5-4： 长导线的磁场

半径为 a 的长（几乎无限）直导线承载均匀分布在其横截面上的稳定电流 I。确定 (a) r ≤ a（导线内部）和 (b) r ≥ a（导线外部）的磁场 H 距导线轴的距离 r。

解：(a) 令电流 I 沿 +z 方向（图5-17(a)）。为求距轴线 r = r_1 ≤ a 处的磁场 H_1，选取半径为 r_1 的圆形安培环路 C_1（图5-17(b)）。安培环路定律给出

$$\oint_{C_1} \mathbf{H}_1 \cdot d\mathbf{l}_1 = I_1, \tag{5.48}$$

其中 I_1 是流过 C_1 的总电流 I 的比例。从对称性来看，H_1 的大小必须恒定，并且沿路径的任何点都与轮廓平行。此外，为了满足右手定则并且假设 I 沿 z 方向，H_1 必须沿 $+\phi$ 方向。因此，$H_1 = \hat{\boldsymbol{\phi}} H_1$、$dl_1 = \hat{\boldsymbol{\phi}} r_1\, d\phi$ 以及等式的左侧： (5.48) 变为

$$\oint_{C_1} \mathbf{H}_1 \cdot d\mathbf{l}_1 = \int_0^{2\pi} H_1 (\hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}}) r_1\, d\phi = 2\pi r_1 H_1.$$

流过 C_1 围成的面积的电流 I_1 等于总电流 I 乘以 C_1 围成的面积与导线总横截面积的比值：

$$I_1 = \left(\frac{\pi r_1^2}{\pi a^2}\right) I = \left(\frac{r_1}{a}\right)^2 I.$$

等式两边相等。 (5.48) 然后求解 H_1 得出

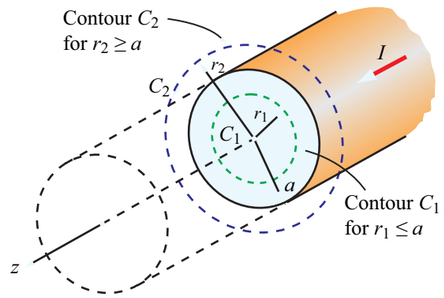
$$\mathbf{H}_1 = \hat{\boldsymbol{\phi}} H_1 = \hat{\boldsymbol{\phi}} \frac{r_1}{2\pi a^2} I \qquad (\text{for } r_1 \le a). \tag{5.49a}$$

(b) 当 r = r_2 ≥ a 时，选取包围全部电流 I 的环路 C_2。H_2 与有向线元均沿方位角方向，故

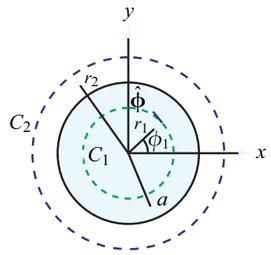
$$\oint_{C_2} \mathbf{H}_2 \cdot d\mathbf{l}_2 = 2\pi r_2 H_2 = I,$$

这产生

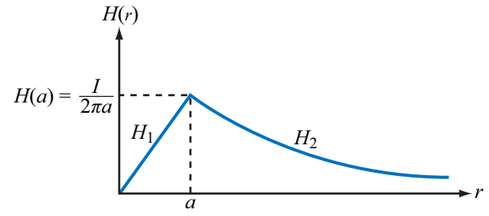
$$\mathbf{H}_2 = \hat{\boldsymbol{\phi}} H_2 = \hat{\boldsymbol{\phi}} \frac{I}{2\pi r_2} \qquad (\text{for } r_2 \ge a). \tag{5.49b}$$



(a) Cylindrical wire



(b) Wire cross section



(c)

图 5-17 半径为 a 的无限长导线，沿 +z 方向承载均匀电流 I： (a) 显示轮廓 C_1 和 C_2 的一般配置； (b) 剖面图； (c) H 与 r 的关系图（示例 5-4）。

忽略下标 2，我们观察到等式： (5.49b) 为 $B = \mu_0 H$ 提供了与等式 1 相同的表达式。 (5.30)，这是根据毕奥-萨伐尔定律推导出来的。

H 的大小作为 r 的函数的变化为

plotted in **Fig. 5-17(c)**; H increases linearly between $r = 0$ and $r = a$ (inside the conductor), and then decreases as $1/r$ for $r > a$ (outside the conductor).

Exercise 5-10: A current I flows in the inner conductor of a long coaxial cable and returns through the outer conductor. What is the magnetic field in the region outside the coaxial cable and why?

Answer: $\mathbf{H} = 0$ outside the coaxial cable because the net current enclosed by an Ampèrian contour enclosing the cable is zero.

Exercise 5-11: The metal niobium becomes a superconductor with zero electrical resistance when it is cooled to below 9 K, but its superconductive behavior ceases when the magnetic flux density at its surface exceeds 0.12 T. Determine the maximum current that a 0.1-mm diameter niobium wire can carry and remain superconductive.

Answer: $I = 30$ A. (See EM.)

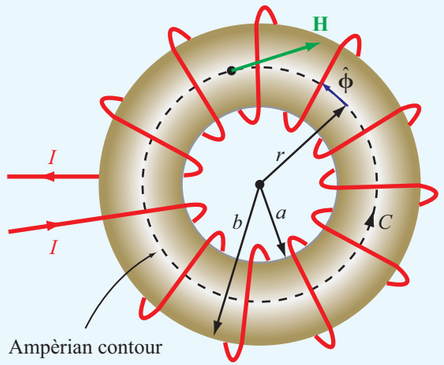


Figure 5-18 Toroidal coil with inner radius a and outer radius b. The wire loops usually are much more closely spaced than shown in the figure (Example 5-5).

Example 5-5: Magnetic Field inside a Toroidal Coil

A ***toroidal coil*** (also called a torus or toroid) is a doughnut-shaped structure (called the core) wrapped in closely spaced turns of wire (**Fig. 5-18**). For clarity, we show the turns in the figure as spaced far apart, but in practice, they are wound in a closely spaced arrangement to form approximately circular loops. The toroid is used to magnetically couple multiple circuits and to measure the magnetic properties of materials, as illustrated later in **Fig. 5-31**. For a toroid with N turns carrying a current I, determine the magnetic field $\mathbf{H}$ in each of the following three regions: $r < a$, $a < r < b$, and $r > b$ with all in the azimuthal symmetry plane of the toroid.

Solution: From symmetry, it is clear that $\mathbf{H}$ is uniform in the azimuthal direction. If we construct a circular Ampèrian contour with center at the origin and radius $r < a$, there will be no current flowing through the surface of the contour. Therefore,

$$\mathbf{H} = 0 \qquad \text{for } r < a.$$

Similarly, for an Ampèrian contour with radius $r > b$, the *net* current flowing through its surface is zero because an equal number of current coils cross the surface in both directions; hence,

$$\mathbf{H} = 0 \quad \text{for } r > b \text{ (region exterior to the toroidal coil).}$$

For the region inside the core, we construct a path of radius r (**Fig. 5-18**). For each loop, we know from Example 5-3 that the field $\mathbf{H}$ at the center of the loop points along the axis of the loop, which in this case is the ϕ direction, and in view of the direction of the current I shown in **Fig. 5-18**, the right-hand rule tells us that $\mathbf{H}$ must be in the $-\phi$ direction. Hence, $\mathbf{H} = -\hat{\boldsymbol{\phi}}H$. The total current crossing the surface of the contour with radius r is NI and its direction is into the page. According to the right-hand rule associated with Ampère's law, the current is positive if it crosses the surface of the contour in the direction of the four fingers of the right hand when the thumb is pointing along the direction of the contour C. Hence, the current through the surface spanned by the contour is $-NI$. Application of Ampère's law then gives

$$\begin{aligned}\oint_C \mathbf{H}\cdot d\mathbf{l} &= \int_0^{2\pi} (-\hat{\boldsymbol{\phi}}H)\cdot\hat{\boldsymbol{\phi}}r\,d\phi \\ &= -2\pi rH \\ &= -NI.\end{aligned}$$

Hence, $H = NI/(2\pi r)$ and

$$\mathbf{H} = -\hat{\boldsymbol{\phi}}H = -\hat{\boldsymbol{\phi}}\frac{NI}{2\pi r} \qquad (\text{for } a < r < b). \tag{5.50}$$

绘制在图5-17(c)中； H 在 r = 0 和 r = a（导体内部）之间线性增加，然后对于 r > a（导体外部）以 1/r 形式减小。

练习 5-10：电流 I 在长同轴电缆的内导体中流动，并通过外导体返回。同轴电缆外部区域的磁场是什么？为什么？

答案：电缆外部 H = 0，因为包围同轴电缆的安培环路所包围的净电流为零。

练习 5-11：当金属铌冷却到 9 K 以下时，它会成为电阻为零的超导体，但当其表面磁通密度超过 0.12 T 时，其超导行为就会停止。确定 0.1 毫米直径的铌丝可以承载并保持超导性的最大电流。

答案：I = 30 A。（见 EM。）

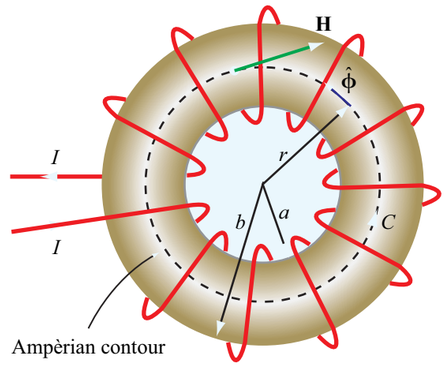


图 5-18 内半径为 a、外半径为 b 的环形线圈。线圈的间距通常比图中所示的要紧密得多（示例 5-5）。

示例 5-5：环形线圈内的磁场

环形线圈（也称为环面或环形线圈）是一种环形结构（称为磁芯），包裹在紧密间隔的线圈中（图 5-18）。为了清楚起见，我们将图中的匝间隔得很远，但实际上，它们以紧密间隔的排列方式缠绕，形成近似圆形的环。环形线圈用于磁耦合多个电路并测量材料的磁性，如后面的图 5-31 所示。对于承载电流 I 的 N 匝环形线圈，确定以下三个区域中每个区域的磁场 H：r < a、a < r < b 和 r > b，所有区域均位于环形线圈的方位角对称平面内。

解：由对称性，H 的大小不随方位角变化。选取以原点为圆心、半径 r < a 的圆形安培环路，则没有电流穿过该环路所围的曲面。因此，

$$\mathbf{H} = 0 \qquad \text{for } r < a.$$

同理，对于半径 r > b 的安培环路，正、反方向穿过所围曲面的电流相互抵消，净电流为零。因此，

H = 0 对于 r > b（环形线圈外部的区域）。

在磁芯内部，选取半径为 r 的圆形环路（图5-18）。由例 5-3，每匝线圈在其中心产生的磁场沿该匝的轴线，即方位角方向。根据图示电流方向及右手定则，H 沿负方位角方向。穿过所围曲面的电流大小为 NI，方向垂直纸面向内。
确定电流正号时，右手四指沿环路 C 的正向弯曲，拇指所指为曲面正法向；据此，本例的有向电流为 −NI。代入安培环路定律得（原书的拇指、四指说明互换，此处已校正）：

$$\begin{aligned}\oint_C \mathbf{H}\cdot d\mathbf{l} &= \int_0^{2\pi} (-\hat{\boldsymbol{\phi}}H)\cdot\hat{\boldsymbol{\phi}}r\,d\phi \\ &= -2\pi rH \\ &= -NI.\end{aligned}$$

因此，H = NI/(2π r) 和

$$\mathbf{H} = -\hat{\boldsymbol{\phi}}H = -\hat{\boldsymbol{\phi}}\frac{NI}{2\pi r} \qquad (\text{for } a < r < b). \tag{5.50}$$